



TNX CODATHON 2K26



Team Name : Fin Fusion

Members Name & Roles:

- Avika Srivastava : UI/UX designer+ Analytics & Deployment
- Ayushi Yadav : AI & Voice integration
- Gauri Mishra : Backend
- Gayatri Bose : Backend
- Harshika Dixit(Leader) : Frontend

Track : Fintech/Artificial Intelligence

VoiceKhata

An AI-Powered Voice-First Digital Ledger for Small Merchants



Problem Statement

The Problem :

Millions of small merchants across India, including kirana stores, tea stalls, pharmacies, and street vendors, still rely on handwritten notebooks (Khata) to record customer credit transactions.

Although digital ledger applications exist, many require typing, reading, and navigating multiple screens, making them difficult for merchants with limited digital literacy.

Challenges Faced :

- Manual bookkeeping is slow and prone to errors.
- Paper records can be lost or damaged.
- Typing transactions during busy business hours is inconvenient.
- Existing apps provide limited support for local languages and natural speech.
- Tracking customer balances becomes difficult over time.
- Anyone can manipulate the ledger entries, which may result in financial losses for the vendor and their business

Our Goal/ Aim of our project :

To build a simple voice-first digital ledger that allows merchants to record transactions naturally by speaking in their preferred language, without typing or complicated interfaces.



Solution Overview

VoiceKhata

VoiceKhata is an AI-powered voice-enabled bookkeeping assistant that converts spoken transactions into organized digital ledger entries.

Instead of typing, the merchant simply says:
"Ramesh ne ₹200 ka samaan udhaar liya "

The application automatically:

- Converts speech into text
- Identifies the customer's name
- Extracts the transaction amount
- Detects whether it is credit or payment
- Updates the customer's ledger instantly
- Recalculates the remaining balance

Key Features

- Voice-based transaction entry
- Automatic customer detection
- Smart ledger management
- Running balance calculation
- Transaction confirmation before saving
- Voice/Text balance enquiry

And all this can be easily done in Hindi-English mixed speech.

Innovation

Unlike traditional bookkeeping apps that rely on typing, VoiceKhata is designed around **natural human conversation**, making digital bookkeeping accessible even for merchants with limited digital skills.

Technology Stack

Frontend



 HTML5
 Tailwind CSS
 JavaScript .

Backend



 Python
 FastAPI

Handles API requests,
transaction processing, and
ledger management.

Speech Recognition



◆ Web Speech API / Whisper

Converts spoken Hindi-
English conversations into
text in real time.

Text Processing



 Rule-Based NLP
 Python Regex

Extracts: Customer
Name, Amount ,
Transaction Type,
Additional Notes, etc
from natural speech.

UI/UX Design



 Figma

Interactive prototype and
user interface design

Development Tools



 Visual Studio Code
 GitHub
 Figma



Implementation Roadmap

1. Problem Analysis

Identify challenges faced by small merchants.

2. Design

Create UI and user flow in Figma.

3. Development

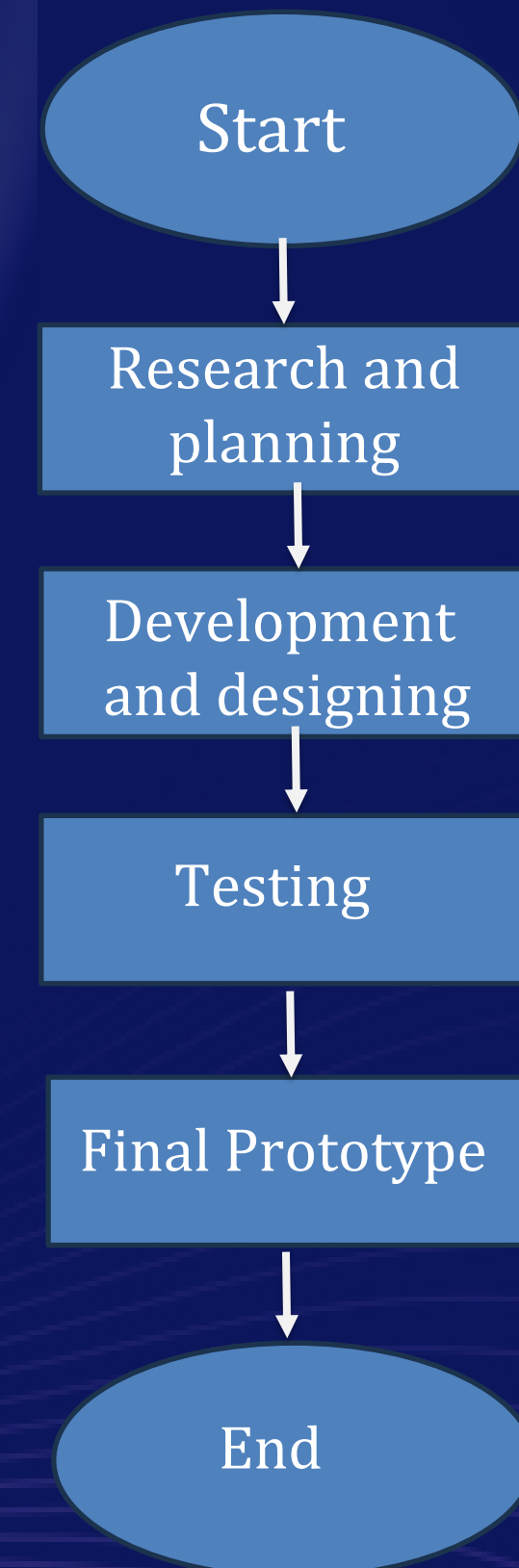
Build the frontend and backend.
Integrate voice recognition.

4. Testing

Test voice input and ledger updates.
Fix bugs and improve accuracy.

5. Final Prototype

Prepare the working demo and presentation.



Expected Impact



Target Users

Kirana Stores
Grocery Shops
Tea Stalls
Pharmacies
Small Retail Businesses
Street Vendors



Business Impact

Faster transaction recording
Reduced manual bookkeeping errors
Easier customer balance management
Improved operational efficiency



Social Impact

- Promotes financial inclusion
- Supports merchants with low digital literacy
- Reduces dependence on paper records
- Encourages digital adoption among small businesses



Expected Outcome

VoiceKhata simplifies bookkeeping by allowing merchants to continue working the way they naturally communicate, making digital finance more accessible and practical.



Future Scope

Although our hackathon focuses on the Minimum Viable Product (MVP), VoiceKhata has strong potential for future expansion.

Planned Enhancements

- Voice-based balance enquiry
- WhatsApp chatbot integration
- Multi-language support
- Smart payment reminders
- Daily transaction summaries
- Analytics dashboard
- Inventory management
- OCR-based bill scanning
- Offline functionality
- UPI payment integration

Long-Term Vision

To become a complete AI-powered financial assistant for India's small merchants, helping them manage bookkeeping, payments, and business insights through simple voice interactions.



Prototype Preview

VoiceKhata

Good day, shopkeeper

H

Your data is synced with the local database.

Overview

Today's sales

₹1,000

Outstanding credit

₹4,150

Customers

5

Add by voice

Speak a new transaction

>

Recent transactions

Reports

Dashboard

Add by voice

Speak a new transaction

>

Recent transactions

Reports

S	Sheela	Goods - 10 Aug	₹2,000
R	Ramesh Kumar	Goods - 10 Aug	₹3,000
R	Ramesh Kumar	Goods - 10 Aug	₹200
S	Sheela	Goods - 9 Aug	+₹4,000
R	Ramesh Kumar	Goods - 9 Aug	+₹200

Customer Transaction/ balance information



Reports

Daily sales

₹1,000

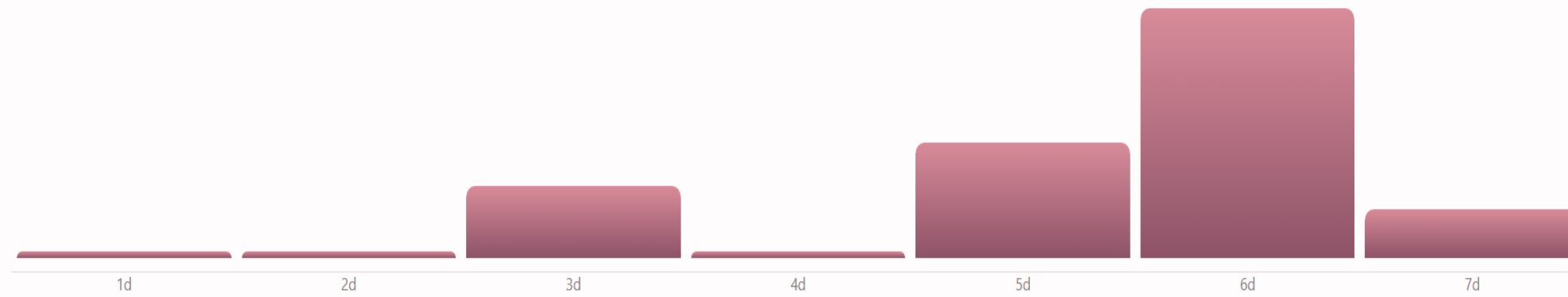
Weekly sales

₹10,100

Total payments

₹5,950

Sales this week



Bars are calculated from saved credit transactions.

Reports and analytics

Tell me what happened

Example: "Ramesh ko 1200 रुपये का सामान उधार दिया."



Heard: सुनीता ने आईएस 2000 का सामान खरीदा

सुनीता ने आईएस 2000 का सामान खरीदा

Extract transaction

Voice enabled bookkeeping